Quantitative Analysis of Media Bias and Stock Price Dynamics: The 2020 Shock

Shivansh Verma
Dept of CS
Ashoka University
Sonipat, India
shivansh.verma_ug2023@ashoka.edu.in

Soham Tulsyan
Dept of CS
Ashoka University
Sonipat, India
soham.tulsyan_ug2023@ashoka.edu.in

Sashwat Dhanuka
Dept of CS
Ashoka University
Sonipat, India
sashwat.dhanuka_ug2023@ashoka.edu.in

Anirban Sen
Dept of CS
Ashoka University
Sonipat, India
anirban.sen@ashoka.edu.in

***Abstract*—**

I. Introduction

II. Related Work

A. Media Sentiment and Market Outcomes

The question of whether the tone of the news moves prices has drawn a wave of recent work, and its trajectory explains where our study sits. Costola et al. [1] ran machine-learning sentiment over COVID-19 news and found that a darker news tone accompanied contemporaneous market reactions, while Bai et al. [2] showed that a market-level sentiment index carries predictive content for returns at short horizons. Both findings live at the level of the whole market, and that is precisely their limit: an aggregate index cannot say whether warm coverage of one firm and hostile coverage of another cancel or compound, which is the distinction a firm-level question needs. The same limit runs through the studies that compare channels and sources. Nyakurukwa and Seetharam [3] find that social-media tone has come to dominate news tone for United States returns, Xu et al. [4] show that official-media sentiment moves the Chinese market, and a broad set of recent papers measures the sentiment–return link across horizons, frequencies, and machine-learning pipelines [5]–[8], across news versus social channels [9]–[11], and across the day–night boundary of the trading calendar [12]. Each of these works reads tone off an aggregate or an outlet; none of them attributes it to the specific firm a headline names, which is the measurement our study is built on.

B. Measuring Tone at the Level of a Firm

Attributing tone to the right entity is its own research problem. A survey of text-based financial sentiment by Todd et al. [13] closes by naming entity- and aspect-level attribution as the open direction of the field, and the tooling has been catching up to that call. Huang et al. [14] show that a finance-domain language model extracts information from financial text more faithfully than general-purpose models, and Hamborg and Donnay [15] supply the piece we lean on directly: a target-dependent stance model that scores the sentiment a sentence expresses toward a named entity rather than the sentence's overall mood. At the same time, recent work counsels care in how such tools are used. Kirtac and Germano show both that language-model sentiment can support profitable trading signals [16] and that the measured sentiment shifts with the choice of model and prompt [17], a sensitivity Liu et al. [18] confirm on S&P 500 headlines; Shah et al. [19] show how weak supervision can stretch scarce labels in finance text, and aspect-level sentiment with large language models remains an active frontier [20], [21]. We take the warning seriously: our pipeline fixes a single supervised relevance stage and a transparent signed score, so that every downstream number traces to one auditable measurement rather than to a moving model choice.

C. Dynamic Links and Structural Change

A third strand asks not whether tone and returns are related but in which direction the relation runs, and whether it survived 2020 intact. Mamaysky [22] dates a structural break in mid-March 2020 after which news and markets became markedly less coupled, which turns the stability of the news–market link from an assumption into something a study must test. On direction, Mukherjee [23] examines sentiment and market movements through Granger causality and Farrell and O'Connor [24] find that the predictive power of a widely watched sentiment index over United States returns comes and goes rather than holding steady; both, again, work with one aggregate series. The COVID-era event studies point the same way from the cross-section: the shock's imprint was uneven across firms and sectors [25]–[29], coverage tone itself has a structural, non-random component [30], and the econometric literature shows that inference which ignores breaks and regime shifts is distorted by them [31], [32]. What no study in this strand does is test the lead–lag relation between firm-level stance and firm-level returns in both directions, on series whose stationarity has been verified and whose break dates are estimated rather than imposed. That combination, firm-level target-dependent stance at scale, level-shift tests for stance and returns under one controlled break, and a break-dated bidirectional dynamic analysis, is the gap this paper fills.

III. Data Collection

A. Corpus and Coverage

Our object of study is the tone of the news that surrounds large United States firms, so the corpus has to begin as close as possible to the full stream of that coverage. We assemble it from MediaCloud, a service that continuously indexes the output of national news outlets, and we query its United States national collection one firm at a time over the period 2015 to 2025. Each query returns the news items that name the firm, each carrying a publication date, a source outlet, and a headline. We keep the headline as the unit of analysis, because the headline is the one part of a news item written to compress the story's angle into a single line, and it is therefore where a firm's coverage wears its stance most plainly. Cast this way the net is deliberately wide: seeded with roughly 250 large-capitalization United States firms, the raw pull returns 6,284,404 headlines, a stream large enough that no single outlet or stretch of time dominates it.

A corpus of this size invites an obvious temptation, which is to make it larger still, and we tested whether a second aggregator, GDELT, could extend the coverage we had already gathered. It could not. The headlines GDELT returned were of visibly poorer quality, often truncated or malformed and heavily duplicated across near-identical records, so folding them into the corpus would have thinned a clean signal with noise rather than deepening it. We keep the study on the MediaCloud stream, whose headlines are consistently well formed and attributed, and treat that stream as the raw material for everything that follows.

One feature of that raw material shapes every later choice. Coverage is spread across firms with extreme inequality: a small number of the most heavily reported names account for the bulk of all headlines, while a long tail of firms surfaces in the news only occasionally. This unevenness is the first thing the data asks us to confront once we begin turning a raw stream of headlines into a panel we can measure.

B. Data Preprocessing

The skew is not a nuisance to be smoothed away; it decides which firms can be studied at all. A daily, firm-level measure of tone is only as steady as the coverage beneath it, and a firm the news names a handful of times a month cannot carry a daily series that means anything. We therefore keep the firms the press reports on heavily and let the thin ones go. Ranking the universe by the volume of coverage each firm attracts and holding onto the most reported names concentrates the corpus onto its densely covered head, some 4,706,589 headlines, and leaves behind a long tail of firms that never generated enough news to measure.

Volume in a single year is not enough on its own, because coverage also has to persist across the whole decade: a firm that falls quiet for years leaves stretches of time that no series can span. We therefore hold each remaining firm to a floor of

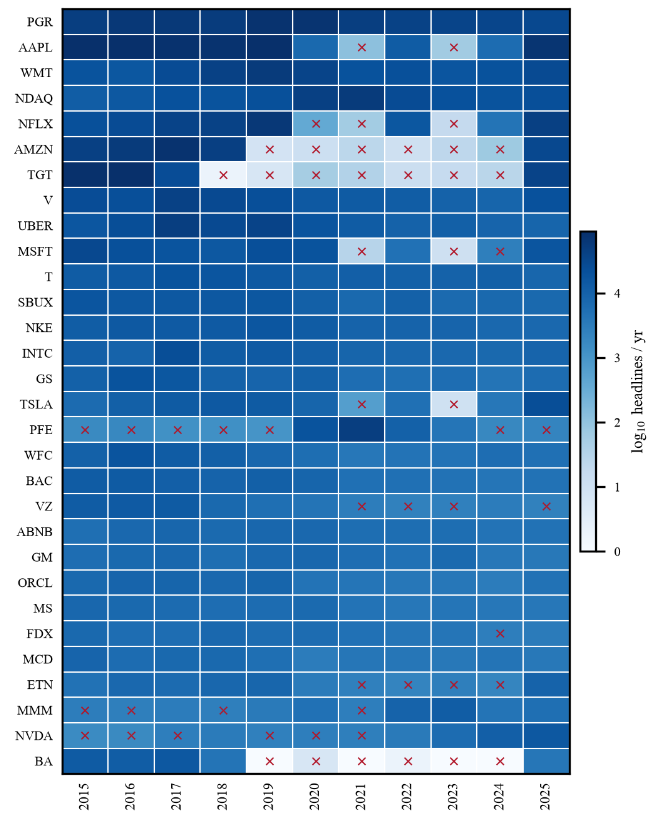

Fig. 1. Yearly headline coverage for each candidate firm, shaded by $\log_{10}$ headlines per year. A cross marks any year in which a firm sits at or below the three-thousand-headline floor; a firm crossed in more than three of the eleven years is set aside as too intermittently covered to support a continuous series.

more than three thousand headlines in a year, and set aside any firm that slips to or below that floor in more than three of the eleven years from 2015 to 2025, since a firm that runs thin that often cannot anchor a continuous weekly or daily series. Figure 1 maps yearly coverage firm by firm and marks the years that fall below the floor; the firms carried forward are those whose rows stay dark almost throughout, while the ones that flicker pale in year after year are the ones the rule removes.

Two kinds of contamination still sit inside what survives. A great many headlines name a firm only in passing, in a list or an aside, while the story itself is about something else; because a headline written about the firm almost always carries the firm's name or its ticker symbol, we keep only the headlines that do and discard the rest. The same wire report, in turn, is republished word for word across dozens of outlets, so we collapse these near-identical copies to a single record per firm. Table I traces the corpus through these passes, from the full scrape down to what remains: 695,731 scored headlines across 26 firms, each attached to a firm the news followed closely and each naming that firm in its title. Even within this retained set coverage stays strikingly unequal, spanning more than four orders of magnitude from the most reported

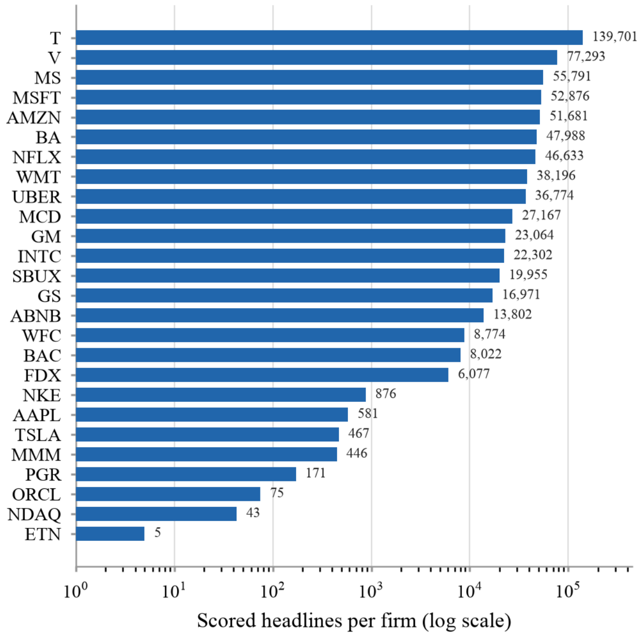

Fig. 2. Scored headlines per firm in the final corpus (26 firms, log scale). Coverage spans more than four orders of magnitude, from AT&T at 139,701 down to Eaton at 5, which the firm fixed effects in the panel regressions absorb.

firm to the least (Figure 2); the estimation later leans on firm-level effects precisely so that these permanent differences in coverage volume cannot masquerade as differences in tone.

TABLE I
FROM THE FULL NEWS SCRAPE TO THE CORPUS THE STUDY RUNS ON.

Stage	Headlines	Firms
Full MediaCloud scrape	6,284,404	250
Most-covered firms retained	4,706,589	30
Name or ticker in title, de-duplicated	695,731	26

IV. METHODOLOGY

A. Relevance Classification

The scored corpus still carries a problem that no amount of name matching can fix: a headline that names a firm is not necessarily about the firm. “Apple” headlines a recipe as easily as an earnings call, banks lend their names to stadiums and sponsorships, and a firm can drift through a story as scenery rather than subject. If such headlines flow into a firm’s daily tone, the signal we later regress on is diluted by coverage that no investor would read as news about the company. What the study needs is the subset of headlines in which the firm is the primary subject of a material corporate event: earnings, mergers and acquisitions, executive changes, large layoffs, regulatory or legal action, major product launches, or credit and bankruptcy events. “Goldman Sachs beats profit estimates on a trading surge” belongs in that subset; “Ten apple recipes to try this autumn” does not, even though both contain a firm name.

Separating the two at the scale of hundreds of thousands of headlines calls for a supervised classifier, and a classifier calls for labels. We hand-labelled 750 headlines against the written rubric above, resolving borderline cases by review, which yielded 227 relevant and 523 irrelevant examples; a further 187 labelled headlines (145 irrelevant, 42 relevant) were locked away as a held-out test set, touched neither during training nor when choosing the decision threshold. The classifier itself is a DeBERTa-v3-base model fine-tuned for binary classification on the headline text: each headline is tokenized to at most 128 tokens, the model produces a 768-dimensional contextual representation, and a linear head maps that representation to a relevance probability. Training uses the AdamW optimizer at a learning rate of 2×10^{-5}, a batch size of 16, weight decay of 0.01, and early stopping on the validation loss.

Trained on the 750 labels alone, the model reaches a precision of only 0.65 at a recall of 0.71, and the reason is instructive: the labelled set is nearly a third relevant while the true rate in the corpus is closer to one headline in eight, so the model learns to say yes too readily. More manual labels would close that gap slowly and expensively. Instead we widen the training set with 10,025 synthetically generated headlines, built from sector-aware templates so that the events, amounts, and counterparties in each generated headline are plausible for the firm’s industry, and aimed squarely at the categories the base model confuses most: consumer promotions, analyst previews, immaterial lawsuits, and passing mentions on the irrelevant side; earnings, layoffs, and deals phrased in unusual ways on the relevant side. We verify samples of the generated headlines against the rubric and confirm that none duplicates the held-out test set. Training on the combined data uses focal loss, which concentrates learning on the hard boundary cases, and weights each manual label five times a synthetic one, so the synthetic examples stretch the decision boundary without ever outvoting a human judgment.

The finished model justifies the effort. On the held-out test set it reaches an accuracy of 0.94 with a precision of 0.88, a recall of 0.86, and an F1 of 0.87 at the default threshold, and the trade-off between catching relevant headlines and keeping false alarms out can be tuned along the threshold sweep in Figure 3. For full-corpus inference we adopt a stricter operating threshold of 0.80, where precision rises to 0.92 at a recall of 0.81 and accuracy holds at 0.94 (34 true positives, 3 false positives, 142 true negatives, 8 false negatives): when the classifier labels the whole corpus, we would rather lose a genuine story than admit a false one, because every admitted headline feeds directly into a firm’s measured tone. Applied to the scored corpus, the classifier retains 90,579 headlines as materially relevant, about 13% of the total, which is consistent with how much of a large firm’s news flow reports corporate substance rather than mentions it in passing. These retained headlines are the raw material for every panel that follows.

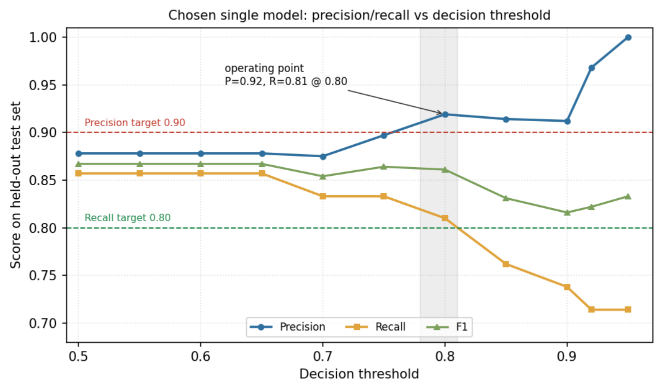

Fig. 3. Precision, recall, and F1 of the final relevance classifier on the held-out test set ($n = 187$) as the decision threshold varies. The operating threshold of 0.80 trades recall for precision ($P = 0.92$, $R = 0.81$), the direction the full-corpus inference favors.

B. A Panel Design for the 2020 Break

With a daily, firm-level stance series in hand and a matching series of daily stock returns, the first question the 2020 shock raises is the plainest one: did either series step to a new level once the shock arrived? The tempting way to answer it is to average each series over the years before 2020 and over the years after, and read off the difference. That answer would mislead, and seeing why fixes the design for both questions that follow.

Two forces move a firm's coverage and its returns that have nothing to do with the shock. Firms differ from one another in ways that hold across the whole sample: some are simply covered more, and more warmly, than others, and some earn higher average returns than others. And on any single day every firm is pushed in the same direction by the news cycle and by the market as a whole. A raw before-and-after average folds both of these into what it labels the "2020 effect": a shift in which firms happen to be in the news, or a run of shared bad days, would surface as a break that owes nothing to the shock itself.

A panel regression with two sets of fixed effects strips out exactly these two confounders. Firm effects pin each firm to its own baseline, so the comparison is between a firm and its own past rather than between one firm and another; time effects hold constant whatever is common to every firm on a given day, or in a given month, so a market-wide swing cannot be read as a firm-level break. What remains for a post-2020 indicator to explain is the within-firm change net of common shocks, which is exactly the quantity the shock question asks about. This is why we estimate the break inside a two-way fixed-effects regression rather than from raw means, from a single firm's time series, or from an event study around one date: only the panel holds the standing differences between firms and the shared movements across them fixed at the same time. The estimate is credible under three assumptions the setting makes reasonable: that, absent the shock, the pre- and post-periods would have shared a common trend; that the timing of the break is set by the pandemic's onset rather than by anything in a firm's own coverage or returns; and that, after the within-transformation, what is left is uncorrelated with the post indicator. To guard the last of these we report HC1-robust standard errors throughout, and, where several firms are covered on the same day, standard errors clustered by day.

C. RQ1: Did Media Stance Shift After the Shock?

The outcome for this question is the tone of a firm's coverage on a day. Each retained headline is read by a target-dependent sentiment model that reports, for the firm named in it, how likely its tone toward that firm is positive, negative, or neutral; write these three probabilities as p_a^+, p_a^-, and p_a^0 for headline a, with $p_a^+ + p_a^- + p_a^0 = 1$. We collapse them into a single signed score, $p_a^+ - p_a^-$, which runs from -1 for wholly unfavorable coverage to $+1$ for wholly favorable coverage and sits near zero when the headline is even-handed or the model is unsure. A firm's stance on a day is the average of these scores over the headlines it drew that day,

$$\mathrm{bias}_{it} = \frac{1}{N_{it}} \sum_{a \in A_{it}} \left(p_a^+ - p_a^-\right), \tag{1}$$

where A_{it} is the set of retained headlines about firm i on day t, $N_{it} = |A_{it}|$ is that day's coverage volume, and $\mathrm{bias}_{it} \in [-1, 1]$ is the average tone toward the firm.

We ask whether this daily stance moved to a new level after the shock. The regression places stance on the post-2020 indicator alongside firm and daily time effects and the day's coverage volume,

$$\mathrm{bias}_{it} = \alpha_i + \delta_t^{\mathrm{day}} + \beta\,\mathrm{Post}_t + \gamma\,\mathrm{vol}_{it} + \varepsilon_{it}. \tag{2}$$

Here $\mathrm{Post}_t = \mathbf{1}[\,t \geq \text{2020-01-01}\,]$ switches on for every day from January 2020 onward, and its coefficient β is the quantity of interest; α_i are the firm effects and δ_t^{day} the daily effects. The control $\mathrm{vol}_{it} = N_{it}$ enters because a daily average tightens mechanically as more headlines are folded into it, so holding the count fixed keeps a genuine change in tone from being mistaken for a change in the sheer amount of coverage. We test

$$H_0^{(1)} : \ \beta = 0 \qquad \text{against} \qquad H_1^{(1)} : \ \beta \neq 0,$$

estimating (2) by the within-transformation with HC1-robust standard errors, and, since several firms can be covered on the same day, reporting day-clustered standard errors alongside them.

D. RQ2: Did Firm-Level Returns Shift After the Shock?

The second question asks the same of prices. The outcome is firm i's daily log return, $\mathrm{return}_{it} = \ln P_{it} - \ln P_{i,t-1}$, where P_{it} is the firm's closing price on day t; we draw the daily closing prices, the S&P 500 index, and the VIX volatility index from `yfinance`. Log returns are additive over time and roughly symmetric, which suits a linear model. Returns carry a large common component—when the market rises, most firms rise with it—so the design must strip that component out before it can see a firm-level break. Monthly time effects absorb the broad market regime, and within each month the S&P 500 return and the VIX account for the day-to-day market swings and the level of volatility. The regression is

$$\text{return}_{it} = \alpha_i + \delta_t^{\text{month}} + \beta\, \text{Post}_t + \gamma_1\, \text{sp500\_ret}_t + \gamma_2\, \text{vix}_t + \varepsilon_{it}, \tag{3}$$

with Post_t as before, α_i the firm effects, δ_t^{month} the monthly effects, and (sp500_ret_t, vix_t) the market controls. We again test the shock,

$$H_0^{(2)}: \ \beta = 0 \qquad \text{against} \qquad H_1^{(2)}: \ \beta \neq 0,$$

estimating (3) by the within-transformation with HC1-robust standard errors. Table II gathers every quantity that enters the two regressions, so that the returns model in particular is read as the post indicator and the market controls acting together rather than any one of them alone.

E. RQ3:

V. Results

A. RQ1: No Detectable Shift in Media Stance

Estimating (2) on the 27,627 firm-day stance observations—26 firms across 3,982 trading days—returns the coefficients in Table III(a). The post-2020 coefficient is $\hat{\beta} = +0.011$ with an HC1 standard error of 0.030 ($p = 0.71$), and clustering by day barely disturbs it (SE $= 0.030$, $p = 0.72$), so the within-day dependence between firms is not hiding a result. We cannot reject $H_0^{(1)}$: once each firm is set against its own history and the day's common movement is held fixed, the tone of coverage does not step to a new level after the shock. The volume control behaves exactly as its purpose demands—its coefficient is -0.0038 ($p < 0.001$), so the days on which a firm is most heavily covered are the days its average tone is pulled back toward neutral, the mechanical averaging effect the control exists to absorb. The within R^2 of 0.002 is what one expects when a single firm's day-to-day tone is dominated by the noise of individual headlines. A null here is not a failure to find something; it says the shock left no common, firm-level mark on tone once the news cycle's own daily swings are taken out.

B. RQ2: No Detectable Shift in Firm-Level Returns

The same estimation on returns, (3) over 20,537 firm-day observations, tells the matching story (Table III(b)). The post-2020 coefficient is all but zero, $\hat{\beta} = -0.0001$ (SE $= 0.0013$, $p = 0.92$): returns show no level break either. What the regression does register is the market itself. The S&P 500 return enters with a coefficient of 1.10 ($p < 0.001$)—the familiar market beta near one—and together with the monthly effects it accounts for about a third of the within-firm variation in returns ($R^2 = 0.32$). Volatility adds nothing once the market return is present (VIX, $p = 0.68$). The reading echoes RQ1: firm returns move with the market, as they always have, but the shock itself leaves no separate firm-level step behind. Taken together the two nulls are informative rather than empty—after common shocks are removed, neither the tone of coverage nor firm returns jumped in 2020. A level test is not the only way the shock could have mattered, though: it could have changed how tone and returns move *with one another* over time rather than where each of them sits, and a static regression cannot see that.

C. RQ3:

VI. Discussion

A. RQ1: What a Null Stance Shift Means

The absence of a detectable stance shift says something sharper than it first appears. Coverage of these firms visibly intensified through the pandemic, yet once each firm is compared with its own history and the shocks common to every firm on a day are held aside, the average direction of that coverage did not lurch. The strong negative volume coefficient completes the picture: what changed in 2020 was how much was written, not how favorably. For an investor or a regulator worried that the crisis bent the press systematically for or against large firms, the daily firm-level evidence here offers no support.

The measurement, however, bounds the claim. A signed daily mean is a coarse summary of tone: it can hide offsetting swings within a day, it inherits noise from both the stance model and the relevance filter, and that noise attenuates any true effect toward zero. A single calendar break cannot separate the pandemic from anything else that changed in early 2020, and headlines, however information-dense, are not the articles beneath them.

The natural extensions follow from those limits. A distributional summary of daily tone would preserve what the mean erases; propagating the stance model's uncertainty into the regression would let the data speak about attenuation rather than leave it as a caveat; and scoring article bodies for the well-covered firms would test how much the headline restriction costs.

B. RQ2: What a Null Return Shift Means

For returns the null carries a different message: once the market factor and the broad regime are absorbed, the 2020 shock left no firm-specific return signature aligned to the break for the average firm in the panel. That is what efficient pricing of a market-wide shock predicts. The pandemic repriced the market as a whole, and it did so through channels the market controls capture, not through a persistent firm-level wedge that survives them.

The design is deliberately conservative, and that conservatism is its limitation. By stripping out the common component, the specification is silent about the large aggregate repricing that plainly occurred; a mean-shift test also cannot see changes in volatility or tail behavior, which moved violently in 2020. The panel treats all firms symmetrically, though the crisis touched an airline and a grocer very differently.

Future work can relax each restriction in turn: modeling conditional volatility directly, defining the firm-specific channel against a richer factor benchmark than the index and the VIX, and replacing the single calendar break with event windows around each firm's own news flow.

TABLE II
VARIABLES IN THE RQ1 (STANCE) AND RQ2 (RETURNS) REGRESSIONS.

Variable	Used in	Meaning
bias_{it}	RQ1 (dep.)	Daily firm stance: the signed mean $p_a^+ - p_a^-$ of the headlines about firm i on day t.
return_{it}	RQ2 (dep.)	Daily log return of firm i, $\ln P_{it} - \ln P_{i,t-1}$.
Post_t	RQ1, RQ2	Shock indicator, $\mathbf{1}[t \geq \text{2020-01-01}]$; its coefficient β is the parameter of interest.
vol_{it}	RQ1	Coverage volume: the number of retained headlines about firm i on day t.
sp500_ret_t	RQ2	Daily S&P 500 index return, controlling for broad market movement.
vix_t	RQ2	Daily VIX level, controlling for market volatility.
α_i	RQ1, RQ2	Firm effects, absorbing permanent differences across firms.
δ_t^{day}	RQ1	Daily effects, absorbing shocks common to all firms on a day.
δ_t^{month}	RQ2	Monthly effects, absorbing shocks common to all firms in a month.

Note: "(dep.)" marks the dependent variable; the remaining rows are regressors or fixed effects.

TABLE III
REGRESSION RESULTS FOR RQ1 (STANCE, EQ. 2) AND RQ2 (RETURNS, EQ. 3). BOTH FAIL TO REJECT THE NULL OF NO POST-2020 LEVEL SHIFT ONCE COMMON DAILY (RQ1) AND MONTHLY (RQ2) SHOCKS ARE ABSORBED.

(a) RQ1 — Daily Stance (firm + day FE, HC1 SE)

Variable	Coef.	SE	p
Post ($\hat{\beta}$)	$+0.01096$	0.02964	0.712
Article volume	-0.00381	0.00049	< 0.001
$n = 27{,}627$; 26 firms; 3,982 days; within $R^2 = 0.0017$ day-clustered $p_{\text{Post}} = 0.717$			

(b) RQ2 — Daily Returns (firm + month FE, HC1 SE)

Variable	Coef.	SE	p
Post ($\hat{\beta}$)	-0.00013	0.00132	0.921
S&P 500 return	$+1.09876$	0.01869	< 0.001
VIX	-0.00003	0.00006	0.684
$n = 20{,}537$; 26 firms; within $R^2 = 0.316$			

C. RQ3: A Link That Is Firm-Specific, Not Market-Wide

The dynamic analysis delivers the paper's central interpretive point. Predictability between stance and returns exists, but it lives in individual firms and tends to switch on only after a firm's own structural break; pooled across the panel, with volatility, the market return, and coverage volume controlled, no market-wide lead–lag channel survives in either direction. A pooled test is exactly the design most able to conjure an aggregate effect, and it declines to; the honest reading is heterogeneity, not absence.

The caveats are those of any reduced-form dynamic analysis. Weekly aggregation, chosen because daily firm-level stance is sparse, may alias away a faster channel through which headlines are priced within days or hours. A single estimated break describes firms with several regime changes only partially. And Granger causality is predictive, not structural: a significant test says past stance forecasts returns, not that stance moves prices.

Extending the analysis to daily or event-time frequency for the densely covered firms, allowing multiple breaks or smoothly switching regimes, and pairing the reduced-form evidence with an identification strategy that can speak to mechanism are the directions we consider most promising.

VII. CONCLUSION

We set out to learn whether the 2020 shock changed what the news says about large United States firms, what their stocks do, and how the two move together. From 6.28 million headlines we distilled a corpus of 90,579 materially relevant, target-scored headlines across 26 S&P 500 firms, and put the shock to three tests: pre/post panel regressions for the level of stance and of returns, and a break-dated weekly dynamic analysis for the link between them. The levels did not move: neither media stance nor firm-specific returns shows a significant aggregate shift once common shocks are absorbed. The link between stance and returns turns out to be real but local, surfacing in individual firms after their own structural breaks while no market-wide channel survives the pooled, controlled test. The 2020 shock, on this evidence, did not rewrite the relationship between the financial press and the market; it left a relationship that was firm-specific before the crisis just as firm-specific after it.

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